

ORIGINAL

NATIONAL UNIVERSITY OF SINGAPORE

EXAMINATION FOR  
(Semester I: 2014/2015)

EE5308R – ANTENNA ENGINEERING

November/December 2014 – Time Allowed: 2 Hours

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INSTRUCTIONS TO CANDIDATES

1. This examination paper contains **FIVE (5)** questions, and comprises **EIGHT (8)** printed pages.
2. Answer any **FOUR (4)** questions totally.
3. Each question carries **TWENTY-FIVE (25)** marks, making a total of **ONE HUNDRED (100)** marks for the entire paper.
4. This is an **OPEN-BOOK** examination.

**FUNDAMENTAL CONSTANTS:**

- $\mu_0$  : free-space permeability =  $4\pi \times 10^{-7}$  H/m  
 $\epsilon_0$  : free-space permittivity =  $8.852 \times 10^{-12}$  F/m  
 $c$  : velocity of light in free space =  $3 \times 10^8$  m/s  
 $\eta_0$  : free-space intrinsic impedance =  $120\pi$   $\Omega$

**Q.1 (Total: 25 marks)**

Choose the best answers for the following questions.

1. When “dBi” is the unit of a maximum gain of an antenna, it suggests that
  - a. A half-wave dipole is used as the reference.
  - b. An isotropic radiator is used as the reference.
  - c. An omni-directional monopole antenna is used as the reference.

(3 marks)
2. If the return loss at the input of the antenna 15 dB, the power reflected from an antenna is:
  - a. 15%;
  - b. 10%;
  - c. 5%;
  - d. 3%.

(3 marks)
3. The electrically small antennas *always* suffer
  - a. Low antenna efficiency and narrow operating bandwidth
  - b. Low antenna radiation resistance and low directivity
  - c. Low antenna gain and narrow operating bandwidth
  - d. Both (b) and (c)

(3 marks)
4. The ultra-high Frequency (UHF) RFID reader antenna is circularly polarized for
  - a. Achieving impedance matching of RFID antenna
  - b. Increasing the reading range of RFID system
  - c. Increasing the possibility to detect RFID tags which are randomly oriented
  - d. All the above

(3 marks)
5. Which antenna will be the MOST possible candidate for a wireless system operating with a high gain over a broad frequency range?
  - a. A half-wave dipole.
  - b. A full-wave loop.
  - c. A log-periodic dipole antenna.
  - d. A Yagi-Uda antenna.

(3 marks)

6. A shorting-pin can be used in an electrically low-profile inverted-L antenna to form an inverted-F antenna for
- further lowering the profile of the antenna.
  - improving the impedance matching of the antenna.
  - shortening the radiating arm of the antenna.
  - enhancing mechanical robustness of the antenna
  - controlling the radiation patterns of the antenna.

(5 marks)

7. The input impedance of an HF RFID tag antenna (coil) is designed for
- matching to a standard feed-line impedance of  $50\ \Omega$ .
  - matching to the output impedance of the RFID tag chip used.
  - matching to the conjugate output impedance of the RFID tag chip used.
  - high radiation efficiency of the coil antenna.
  - high gain of the coil antenna

(5 marks)

**Q.2 (Total 25 marks)**

A linear array consisting of three isotropic antenna sources is positioned along the z-axis. The centre antenna is located at the origin. The current excitation for the centre antenna has a magnitude of 4 (A) and for the adjacent element the magnitude is 2 (A). The uniform antenna spacing is one wavelength apart.

- a. Derive and simplify the expression of the array factor for this array. (8 marks)
- b. Find the angle (in degree) for all the possible nulls in the array factor (8 marks)
- c. Compute the angles of all the maxima in the array factor (9 marks)

**Q.3 (Total: 25 marks)**

A linear array of  $N$  isotropic sources is placed along the  $z$ -axis, where the first source of the array is located at the origin of the Cartesian coordinates. The  $N^{\text{th}}$  antenna source is located on the positive  $z$ -axis. The magnitude of each excitation for the element is the same at unity. The uniform spacing of between adjacent antennas is  $d$ .

- a. Derive and simplify the expression of the array factor for  $N=5$ . (8 marks)
- b. Compute the phase excitation for an end-fire array with the main beam pointing towards the positive  $z$ -axis. The centre of the array is now changed to the origin and the spacing between adjacent antenna is quarter of a wavelength. Where are the nulls located? (9 marks)
- c. If  $N=10$  and the antenna spacing is quarter of wavelength, compute the phase excitation required of the uniform array when the main lobe is pointed towards  $45^\circ$  and comment on the characteristics on the pattern of the main lobe.

(8 marks)

**Q.4 (Total 25 marks)**

Q.4a: The radiation resistance of a dipole is 40 ohm while the ohmic loss caused by the materials of the dipole is 3 ohm. The dipole antenna is matched to a 50-ohm feeding line by using an impedance matching network between the dipole and feeding line. The impedance matching network introduces the additional ohmic loss of 5 ohm. The return loss at the interface between impedance matching network and the input of the antenna is ignored.

**(Note: To write all solutions in your Answer Booklet)**

- a. The radiation efficiency of the antenna is \_\_\_\_\_.
- b. The antenna efficiency is \_\_\_\_\_.
- c. If the dipole achieves the directivity of 2 dBi, the gain of the dipole antenna is \_\_\_\_\_ dBi

(3×3=9 marks)

Q.4b: (Select **only one correct answer** a or b for each question)

- RFID systems can operate at both near-field and far-field modes. In the near-field RFID systems, the reader and tags communicate through (1) a. magnetic field coupling or b. scattering of (2) a. the coil-type or b. the dipole-type antennas while (3) a. the coil-type or b. the dipole-type antennas used in far-field RFID tags communicate with the reader antenna through (4) a. the magnetic field coupling or b. the scattering from the tag antennas.
- In the near-field HF RFID systems operating at 13.56 MHz, the impedance of tag antenna and the microchip are respectively  $Z_{\text{tag}}$  and  $Z_{\text{chip}}$ . If the overall output capacitance of a tag microchip is  $C_{\text{chip}}=3.912 \times 10^{-11}$ , the coil of the tag antenna should be designed with (5) a. a capacitive or b. an inductive reactance for (6) a. a maximum reading range of the RFID system or b. the maximum radar cross section (RCS) of the antenna.
- In the far-field UHF RFID systems, the maximum reading range is determined by (7) a. power charging link or b. communication link between the tag and reader antennas. To achieve maximum reading range, the tag antennas should be conjugately matched to the output impedance of (8) a. the tag microchip or b. a standard impedance such as 50 ohm.

(2×8=16 marks)

**Q.5 (Total 25 marks)**

Q.5a. A log-periodic dipole antenna (Figure Q.5a) with optimum parameters is designed to cover the frequency range of 47–80.6 MHz. Suppose the desired gain is 9dBi. Ignore the variation of feed gap,  $s_n$ .

1. To determine the spacing and scaling factors for the optimum design,  $\sigma$  and  $\tau$  as well as the minimum number of the dipoles,  $N$  (2×3=6 marks)
2. To tabulate the lengths of all dipoles,  $L_n$  in meter and all the element spacing,  $d_n$  in meter (4 marks)

**Hint:** to determine optimum design factors from the attached figure.

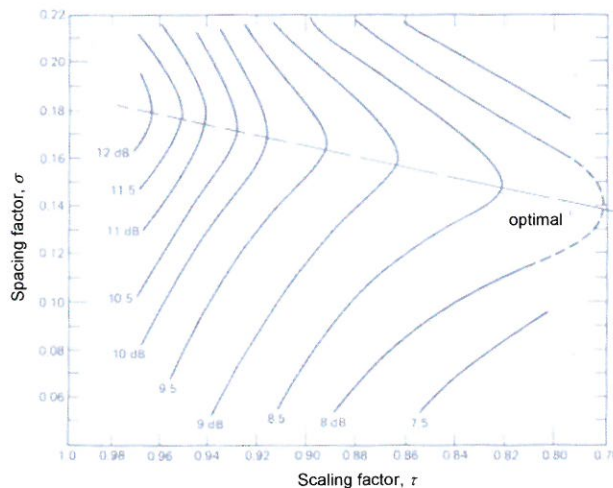
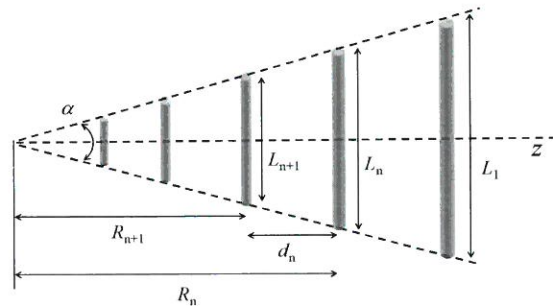


Figure Q.5a

Table Q.5a List of  $L_n$  and  $d_n$

(Note: To write all solutions in your Answer Booklet)

| n               | 1 | 2 | 3 | 4 | 5 |
|-----------------|---|---|---|---|---|
| $L_n(\text{m})$ |   |   |   |   |   |
| $d_n(\text{m})$ |   |   |   |   |   |

Q.5 is continued on Page 8

**Q.5:**

Q.5b. The monopole antenna consists of a straight wire perpendicular to a perfectly electrically conducting plane (ground plane) as shown in Figure Q5b.

1. Draw the equivalent dipole of the original monopole antenna using Image Theory and indicate the source voltages and currents of both original monopole and its image using  $V_{in}$  and  $I_{in}$ .
2. Show that the gain of the original monopole is twice that of the corresponding dipole and that radiation resistance is one-half.

(5+10 marks)

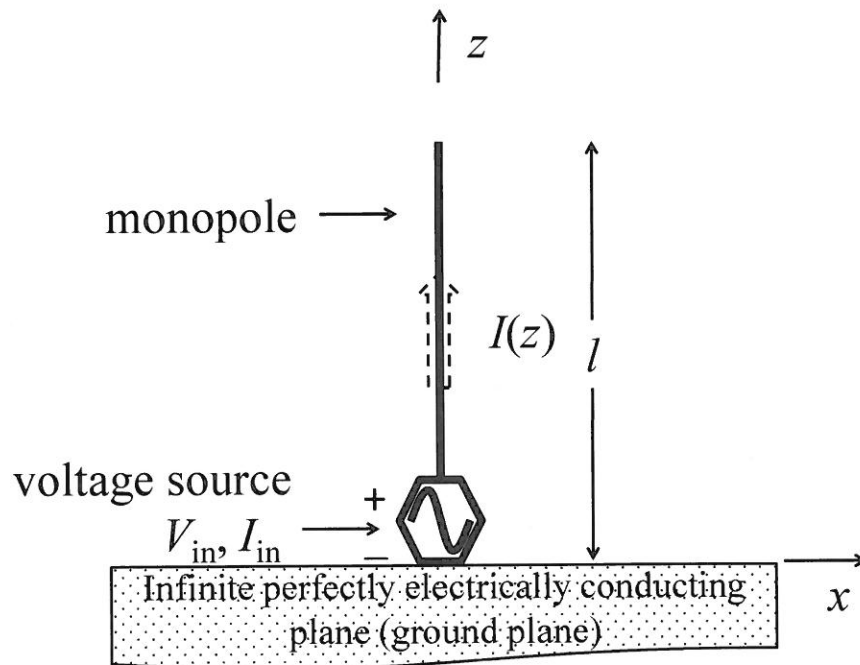


Figure Q5b